OE02-SAMPLES+ CUSTOM FIRMWARE

VERSION 1.2.11 · SUPPLEMENT TO THE LXR OWNERS MANUAL · FOR THE LXR-02

SAMPLES & WAVES

31 WAV samples and 8 single-cycle waves from the SD card, as oscillator and LFO waveforms, with start, length and loops.

SEQUENCER

A step grid per track, ratchets and roll locks, trigger tracks, LFO start, lock trigger and instant pattern switching.

EFFECTS

A second FX slot, sends, pre-fader option and a master bus on St1: saturation, compressor and EQ.

MORPH & MORE

Kit morph per voice and from the mod wheel, voice morph, choke group, voice INIT templates.

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github.com/shellbeat-jpg/lxr-desktop-sample
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TABLE OF CONTENTS

3 **1 INTRO**
3 **1.1.** Reading the tables
3 **1.2.** What is stored where
3 **1.3.** Installing
3 **1.4.** Compatibility

4 **2 MEMORY MANAGEMENT**
4 **2.1.** SD card
4 **2.2.** Sample import
5 **2.3.** Experimental: 1.5 MB sample memory
5 **2.4.** Kits
5 **2.5.** Voice INIT templates

6 **3 VOICES**
6 **3.1.** Engines per voice
6 **3.2.** Oscillator sources
6 **3.3.** Oscillator page (OSC)
7 **3.4.** Samples and loops
7 **3.5.** Waves
7 **3.6.** Modulated sample selection
8 **3.7.** Amplitude envelope page (AEG)
8 **3.8.** Mixer page (MIX)
8 **3.9.** LFO page (LFO)
9 **3.10.** Track 7
9 **3.11.** LFO 2 (test)

9 **4 SEQUENCER**
9 **4.1.** Lock trigger: velocity 0
9 **4.2.** Step flam: ratchets and roll locks
10 **4.3.** Step grid per track
10 **4.4.** Instant pattern switching
10 **4.5.** Trigger tracks
11 **4.6.** LFO start

12 **5 PERFORMANCE MODE**
12 **5.1.** Performance menu
12 **5.2.** Kit morph
12 **5.3.** Voice morph

13 **6 EFFECTS**
13 **6.1.** Signal flow
14 **6.2.** FX1 and FX2
15 **6.3.** Sends and pre-fader
15 **6.4.** FX knobs
16 **6.5.** Master bus FX3

18 **7 GLOBAL SETTINGS**

18 **8 MIDI**

19 **A WHERE SETTINGS ARE STORED**

1 > INTRO

This supplement describes what the firmware **OE02-Samples+ 1.2.11** adds to the LXR-02 and to its owners manual. Everything that is not mentioned here works as described in the manual.

OE02-Samples+ is built on **OpenElixir-02 1.2** by Moritz Waser. Features that come from OpenElixir-02 are marked [OE]; everything else is new in OE02-Samples+. The chapters follow the order of the manual: memory, voices, sequencer, performance mode, effects, global settings and MIDI.

1.1. READING THE TABLES

As in the manual, the parameter tables list the **displayed name** (the three letters on the display), the full name and what the parameter does. The edit view (click the encoder) shows the full name and the value with its unit.

Button combinations are written with the label printed below the step button, for example **SHIFT + FX** (step button 15) or **SHIFT + LAST STEP** (step button 14).

1.2. WHAT IS STORED WHERE

Kit: the sound of the 6 voices (SAVE → Kit). **Pattern**: sequencer data. **Global**: kept after a restart only when saved with SAVE → Global. The appendix lists every new setting with its place.

1.3. INSTALLING

1. Copy **LXR_OE02-Samples_V1.2.11.img** to the top folder of the SD card and delete any other **LXR*.img** files there.
2. Switch the LXR off, insert the card, hold the encoder button and switch it on; keep holding it until "LXR Updater" appears.
3. Wait for "Update OK!", then remove the image from the card.
4. Keep your WAV files in **SAMPLES** and switch on normally: the update erased the samples, they are imported again at the first start.

After switching off, wait about 10 seconds before switching on again; otherwise the display can stay dark (the unit still works).

1.4. COMPATIBILITY

Kits and patterns of the stock firmware and of OpenElixir-02 load unchanged; the new settings start at their defaults (no samples, sends 0, FX3 off, all tracks in 1/16).

Kits saved with 1.2.10 or later load in older firmwares; those ignore what they don't know. Voices with a sample or wave as waveform stay silent there.

To go back, install the OpenElixir-02 or the Erica Synths image the same way.

2 > MEMORY MANAGEMENT

2.1. SD CARD

Besides the projects, the card holds the audio files and two template files:

Folder / file	Content
SAMPLES	one-shot samples, WAV
SAMPLES/LOOPS	loop samples, WAV
SAMPLES/WAVES	up to 8 single-cycle waves, WAV
BIGFLASH	a file or folder with this name in **SAMPLES** switches on the experimental 1.5 MB sample memory (chapter 2.3)
INIT.SND	voice INIT template: in the top folder (global) and in a project folder (chapter 2.5)

2.2. SAMPLE IMPORT

At a normal start the LXR compares the folders with the stored samples (names and sizes). If something changed, it imports again: "Import x/y", then "Import+Verify OK". The samples are stored in the internal flash and also play without a card.

Rule	
Format	WAV, 16 bit PCM, mono or stereo (mixed to mono), any sample rate
Names	8.3 format (**01KICK.WAV**); the slots **S1** ... **S31** follow the order of the names, numbers by value (**2KICK** before **10KICK**)
Number	up to 31 samples (both folders together) and 8 waves
Length	about 5.9 s mono at 44.1 kHz in total (11.9 s at 22.05 kHz); a stereo file counts half its size
Ignored	names starting with **.** or **_**

Files that are not 16 bit are skipped ("Not 16bit, skip:"). Files that don't fit are skipped in name order ("Full(5.9s) skip:"). A file replaced by another one with the same name and size is not imported again: rename it.

START SCREEN

Samples:12 W:3 = stored samples and waves; below **Used 4.2/5.9s** = used and available sample time.

A firmware update erases the samples. Keep **SAMPLES** on the card; the next start imports them again.

2.3. EXPERIMENTAL: 1.5 MB SAMPLE MEMORY

The chip of the LXR-02 reports 1 MB of flash; it usually has more. With **BIGFLASH** in **SAMPLES** the import uses the area above 1 MB as well: about **17.8 s** instead of 5.9 s.

- ST doesn't guarantee that area. Every sample is verified after writing; on an error the LXR shows "Big flash failed / using 512K" and imports into the normal memory.
- The start shows "Big flash: on / 1.5 MB, testing", then "Used x.x/17.8s".
- If the unit ever hangs at the start screen, remove **BIGFLASH** (or start without card). Firmware and bootloader are not affected.

2.4. KITS

A kit now also stores, per voice: the engine (0E), the sample start and length, the choke group and the FX sends. Per kit: FX1, the FX slot order and the master bus FX3. The additions are a small block at the end of the kit file.

2.5. VOICE INIT TEMPLATES

Hold **LOAD** and press the step button of a voice (1–7). From kit 0 turn the encoder one step back: the row **INIT: Project**, **INIT: Global** or **INIT: Factory** shows where this voice's INIT comes from. Click it for the actions:

Action	Description
Load INIT	loads the voice from the project template, else from the global template, else the factory values of its engine
Save to Project	stores this voice of the active kit in the template of the project
Save to Global	stores this voice in the global template
Del Project INIT, Del Global INIT	removes this voice from the template (click again to confirm)
Back	back to the kit list

A template keeps every voice separately, with everything of the voice: engine, sound, LFO and velocity settings, sample, start and length. A project template needs a saved project.

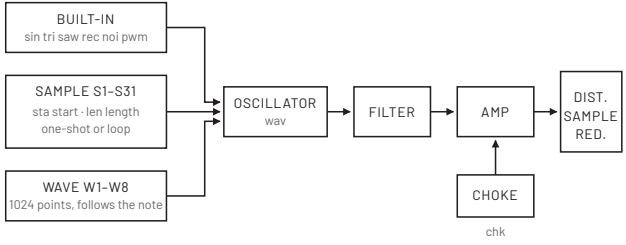
3 > VOICES

3.1. ENGINES PER VOICE OE

Unlike the stock firmware, the type of a voice can be changed: each of the 6 voice slots runs the drum, snare, cymbal or hi-hat engine. **SHIFT + KIT** opens the engine menu of the active voice. The engine is saved in the kit.

3.2. OSCILLATOR SOURCES

The main oscillator of every engine plays the built-in waveforms, a **sample** or a **wave**. Samples play once (or loop) per trigger at their own pitch; waves are single cycles that follow the note like the built-in waveforms.



Shown for the drum voice; the other engines have the same oscillator choice.

3.3. OSCILLATOR PAGE (OSC)

New values and parameters on the OSC page:

Displayed name	Name	Description
wav	Waveform	after **sin tri saw rec noi pwm**: the samples **S1** … **S31**, then the waves **W1** … **W8**. Unused sample slots are skipped when turning. The edit view shows the file name, **L** for a loop and the slot.
sta	Start	in place of **pwm** while a sample is selected: where the sample starts, in ms (0 = beginning).
len	Length	drum 1–3, 5th parameter: how long the sample plays, in ms (0 = **all**, to the end).
coa	Coarse tune	with a sample: 60 = original pitch.

sta and **len** run on a quadratic scale: fine steps at the beginning. The start is taken at the trigger, also with LFO or velocity modulation. Short fades avoid clicks when the start or the length cut into the sample.

The FM page of the drum voices can use samples and waves as FM source as well.

3.4. SAMPLES AND LOOPS

A sample from **SAMPLES** plays once per trigger; the amp envelope (AEG decay) can end it earlier. A retrigger fades the sound that still plays (about 1 ms) instead of clicking.

A sample from **SAMPLES/LOOPS** plays from **sta** to the end set by **len** and jumps back to **sta**, until the amp envelope ends the sound. Drum voices have no gate, so a long decay keeps the loop going; with decay 127 it never ends by itself. Stopping the sequencer fades out every playing loop. The loop point is crossfaded (up to 256 frames, at most a quarter of the loop).

3.5. WAVES

Up to 8 WAV files in **SAMPLES/WAVES**, each one cycle of a waveform. At the import every file is resampled to 1024 points; the end leads back to the start. Stereo files use the left channel only.

- Waves are numbered on their own: **W1** = the first name. Adding or removing samples doesn't move them.
- Best results: one cycle that starts and ends at a zero crossing, rising in the same direction, without DC offset.
- 2 KB per wave in the sample memory.

3.6. MODULATED SAMPLE SELECTION

With a sample as waveform, modulating the waveform (LFO or velocity, target **wav**) selects among the following samples instead of switching to the built-in waveforms:

- The range starts at the selected sample and covers as many further samples as the modulation amount (LFO **mod**, velocity **amt**): S5 with amount 4 = S5 … S9.
- The sample is chosen at the trigger and kept until the next one.
- **Velocity layers**: soft to loud versions named in a row, the softest selected, amount = number of layers – 1.
- **Round robin**: LFO waveform **cyc** steps through the range, one sample per trigger. Random: **rnd** with a rate faster than the hits.

The same works with a wave as base: it selects among the waves.

The three waveform targets of the LFO and velocity menus are shown as **wav** (main oscillator), **fwv** (FM / mod oscillator) and **cwv** (transient); only **wav** selects samples.

3.7. AMPLITUDE ENVELOPE PAGE (AEG)

New on the AEG page of every voice:

Displayed name	Name	Description
chk	Choke	on/off: puts this track into the choke group of the kit. The edit view shows all tracks of the group, e.g. **1—4-6-**.
sna	FX1 send	send into FX slot A (FX1), 0 ... 127, besides the voice's output.
snb	FX2 send	send into FX slot B (FX2), 0 ... 127.

CHOKE GROUP

A track of the group cuts the other tracks of the group when it is triggered, with a fade of about 2 ms; a choked track stays silent until its next trigger. Use it for hi-hat samples on several voices, or an open hat cut by a closed one. Tracks 6 and 7 share one voice and cut each other anyway.

SENDS

The sends are taken after the voice's pan, and after its volume fader unless **pre** is on (chapter 6.3). Set the mix of the slot fully wet for a classic send/return.

3.8. MIXER PAGE (MIX)

Displayed name	Name	Description
out	Audio out	**St1**, **St2**, **L1**, **R1**, **L2**, **R2** as before; **FX** = 100 % into FX slot A; new **FXB** = 100 % into FX slot B.

With **out** = **FX** / **FXB**, a send into the other slot feeds both slots.

3.9. LFO PAGE (LFO)

New waveforms of **wav**:

Value	Waveform
cyc	step counter: one step per trigger; **mod** = number of steps (on **wav**: round robin)
xtr	exponential triangle
si1 tr1 su1 sd1 sq1	one-shot sine, triangle, saw up, saw down, square (a gate)
xu1 xd1 xt1	one-shot exponential up, down, triangle
rn1	a new random value at every trigger
w1 ... w8	the waves, running
w1. ... w8.	the waves, one-shot

ONE-SHOTS

One-shot waveforms run one cycle per trigger and then hold their end value. They start with a trigger of the voice set with **rtg**; with **rtg** off with a trigger of the LFO's own voice. **frq** and **snc** set the length of the cycle.

Displayed name	Name	Description
ofs	Offset	with a one-shot: the delay before the cycle starts, as a part of one cycle (64 ≈ half a cycle).
mod	Amount	the edit view shows the range the target moves in, e.g. **21..127** (target value ± amount).
dst	Destination	an LFO can now modulate rate, amount … of **every** other LFO (the stock firmware: only of LFOs with a higher number).

3.10. TRACK 7

Tracks 6 and 7 play one voice; track 7 has its own decay and its own **wav** (waveform, sample or wave, e.g. the open hat), **coa** (−24 … +24 semitones on top of track 6), **sna**/**snb** and **pan**. Its pages show them under these names in place of the shared ones (edit view: "Trk7 …"; **=t6** = as track 6). The voice uses the sends and pan of the track that triggered last.

3.11. LFO 2 (TEST)

Press the LFO button again on the second LFO screen: **LFO 2** of the voice, same parameters; once more back to the LFO. On track 7 the LFO page is the LFO of track 7: with a target it acts while track 7 plays, instead of the LFO of voice 6. Kit data, not morphed.

4 > SEQUENCER

4.1. LOCK TRIGGER: VELOCITY 0

A step with velocity 0 doesn't trigger the voice and sends no MIDI note; its automation (**P1**/**P2**) is still applied, like a "trigless lock".

4.2. STEP FLAM: RATCHETS AND ROLL LOCKS

New values of the step parameter **flm** (step mode), after **x2** … **x8**:

Value	Effect
x12, x24	12 or 24 repeats within the step
x4+, x4-	4 repeats with rising / falling velocity (25 … 100 % of the step's velocity)
1/8 1/4 1/2 1/1	roll lock: the step repeats every 2, 4, 8, 16 steps of its track until the next active step

4.3. STEP GRID PER TRACK

Each track can run in its own grid, e.g. hi-hats in triplets against straight drums. **SHIFT + LAST STEP** shows **[16] 1/16 1** = length, grid and step length compared with 1/16. **Click the encoder** to edit the grid. In the ALL mode (press LAST STEP again) it sets all tracks.

Grid	Step length (1/16 = 1) · steps per bar
1/4 · 1/4T · 1/8	4 · 8/3 · 2 — 4 · 6 · 8 steps
1/16. · 1/8T · 5/64	6/4 · 4/3 · 5/4 — 10⅔ · 12 · 12⅘ steps
1/16	1 — 16 steps (default)
1/32. · 1/16T	3/4 · 2/3 — 21⅓ · 24 steps
1/32 · 1/32T	1/2 · 1/3 — 32 · 48 steps

Each track loops at its own length (polymeter); the pattern ends after its longest track. The 1/16 tracks keep their timing. Roll (**rol**) and flam repeats count in the track's own steps; live recording is quantized to 1/16. The grid is stored in the pattern.

4.4. INSTANT PATTERN SWITCHING

Global settings, second page, **ips** on: a pattern selected by hand starts on the **next step at the same position** instead of at the end of the playing pattern. Chains and songs still change at the end.

4.5. TRIGGER TRACKS

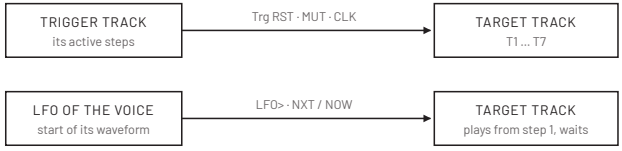
Any track can also act on another track. On the LAST STEP page, every click of the encoder moves on:

Click	The encoder edits
1	length of the track
2	grid
3	trigger mode **Trg [OFF]**
4	its target track **>[T3]**
5	LFO start target **LFO>[T3]**
6	LFO start timing **NXT** / **NOW**

Another click returns to the length. Targets that already have a source are skipped.

Trg	Each active step of the trigger track …
OFF	nothing (default)
RST	sets the target back to its step 1, which plays at once; otherwise the target runs as usual
MUT	mutes the target for as long as the step lasts
CLK	moves the target on by one step; the target only steps on these impulses

Steps of any velocity count; with velocity 0 the trigger step makes no sound itself. The grids of both tracks are kept. A track can be the target of one source only.



4.6. LFO START

LFO>[T3]: the LFO of this track's voice (track 7: voice 6) starts track 3. Each start of its waveform — a new cycle, a restart by **rtg**, a one-shot after its delay — starts the target at step 1:

- **NXT** on the next step of the target's grid (stays in time),
- **NOW** at once.

The target plays to its last step and then waits for the next start. If the LFO cycle is shorter, the next start comes first and loops the part up to there. Until the first start the target is silent.

With **snc** the starts fall on the beat; free-running LFOs drift against the pattern, which can be used for generative restarts. Chaining LFOs (one varies the rate of the next) makes the restarts irregular.

A track can have a trigger target and an LFO target at the same time.

5 > PERFORMANCE MODE

5.1. PERFORMANCE MENU

Turn the encoder on to reach the second page.

Displayed name	Name	Description
bpm rol mrp srt	Tempo, roll, morph, sample rate	as before; **rol** counts in the track's own steps
dj	DJ filter cutoff OE	LP/HP filter on the final mix of all outputs
res	DJ filter resonance OE	
sat	Saturation	of the master bus FX3 (chapter 6.5)
frq	Saturation split	of FX3
shu	Shuffle	as before (second page)
vmv vmo vsw	Voice morph	chapter 5.3

5.2. KIT MORPH

- **Mod wheel**: MIDI CC 1 on the global channel sets **mrp**.
- **Morph target**: **mrp** 0 → 127 morphs from the value in the active kit to the value in the morph kit. In the edit view of a sound parameter hold **SHIFT**: the display shows **Morph>** and the target value, which the encoder changes. **SHIFT + encoder click** sets it to the active value: the parameter no longer morphs.
- **mrv** (global settings, second page): kit morph on/off per voice.

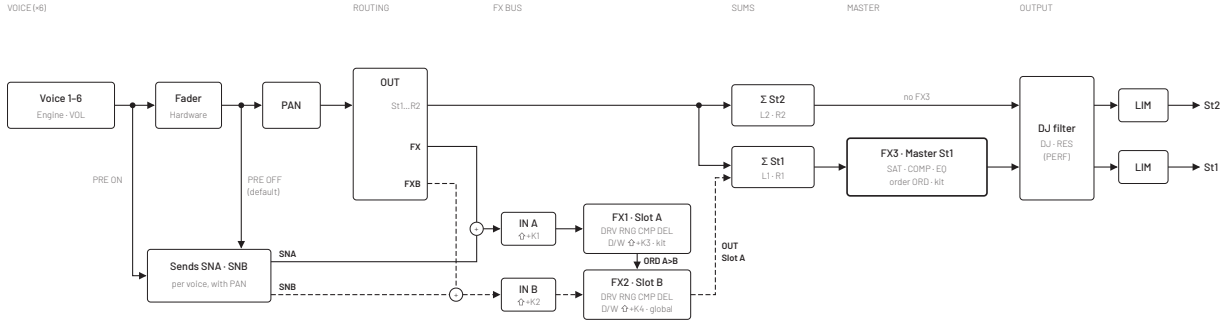
5.3. VOICE MORPH

Displayed name	Name	Description
vmv	Voices	the pair: OFF, 1>2 … 6>5 (source > target)
vmo	Amount	0 = the source as set, 127 = the target's values
vsw	Switch	per source voice: from this **vmo** on, switch-like values (waveform, filter type) are the target's

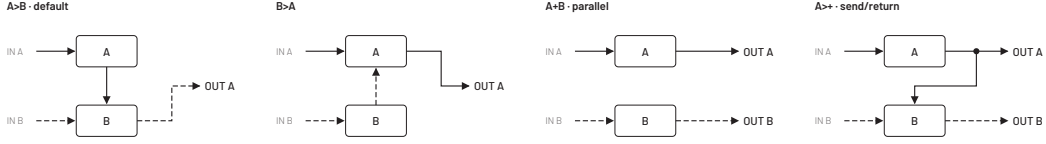
Only **vsw** is kept (global); **vmv** and **vmo** are not saved.

6 > EFFECTS

6.1. SIGNAL FLOW



From the voice to the jacks, slot order A>B (dashed: slot B; ⇧ = SHIFT, K1 ... K4 = parameter knobs). The fader acts on the sends only with **pre** off. FX3 sits on St1 (without a cable in St1 it follows St1 to the St2 jacks).



The order of the FX slots, **ord** on the FX1 page (chapter 6.2).

6.2. FX1 AND FX2

SHIFT + FX shows the FX of the kit (**FX1**, slot A). Press **SHIFT + FX** again for the project FX (**FX2**, slot B), once more for the master bus (**FX3**), then back to FX1. Both slots have the same effects **off drv rng cmp del** and parameters.

Displayed name	Name	Description
typ / fx2	Type	the effect of slot A / slot B
out	Audio out	output of the slot
d/w	Dry/Wet	mix; also SHIFT + knob 3 / 4

Turn the encoder past the last effect parameter for the setup page:

Displayed name	Name	Description
sya / syb	Sync A / B	delay time in note values: OFF, 1/4, 1/8., 1/4T, 1/8, 1/16., 1/8T, 5/64, 1/16, 1/32., 1/16T, 1/32, 1/32T (at most about 0.74 s)
ord	Order	FX1 page: how the slots are connected

ord	Connection
A>B	slot A into slot B, the chain to the output of slot A (default). Slot B is an insert: at 100 % wet the direct sound of A is gone.
B>A	slot B into slot A, to the output of slot A
A+B	side by side, each slot to its own output
A>+	send/return: slot A to its output and into slot B, slot B to its own output

The four connections are drawn on the previous page. FX1 and **ord** are stored in the kit, FX2 and the sync with the global settings.

6.3. SENDS AND PRE-FADER

Each voice sends into slot A with **sna** and into slot B with **snb** (AEG page), besides its own output. Global settings, second page:

Displayed name	Name	Description
pre	Pre-fader sends	**off** (default): the sends come after the voice's volume fader. **on**: before it — the fader only sets the dry signal, the voice stays in the effects (fader down = only the echo is left).

vol on the mix page acts on both in either mode. Voices with **out** = **FX** / **FXB** keep their fader.

6.4. FX KNOBS

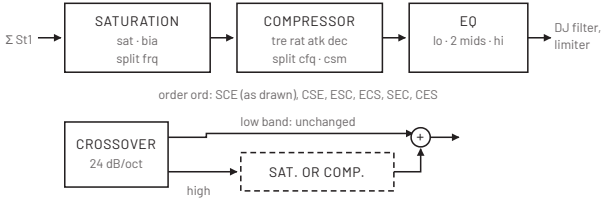
On every page except LOAD/SAVE: hold **SHIFT** and turn a parameter knob.

Knob	Function
1	input gain of FX1 (slot A)
2	input gain of FX2 (slot B)
3	mix (**d/w**) of FX1
4	mix (**d/w**) of FX2

The display shows the values for 1.5 s: **In dB A +0 B -6** / **Mix % A 50 B100**. The input gain scales everything that goes into the slot (outputs FX/FXB and all sends), −∞ … +12 dB, centre 0 dB; it is a global setting. With **pre** on, input gain and mix set the effect levels.

6.5. MASTER BUS FX3

A third FX slot, fixed at the end of the chain on **St1**: after FX1/FX2, before the DJ filter and the limiter. It has three stages — saturation (S), compressor (C) and EQ (E) — in the order set with **ord**. FX3 is stored in the kit; older kits load with FX3 off.



The split: only the band above the crossover is saturated or compressed. **csm** = sc uses the band above only to drive the compressor and compresses the whole signal.

FX3, PAGE 1

Displayed name	Name	Description
sat	Saturation	0 = off; soft clipping, up to +24 dB into the curve, level compensated
frq	Sat split	off = full band, else 40 Hz … 10 kHz: only above it is saturated
cfq	Comp split	off = full band, else 40 Hz … 10 kHz
csm	CFQ mode	**spl** = band split (only the band above is compressed), **sc** = sidechain filter (the band above drives it)
tre	Threshold	−48 … 0 dB
rat	Ratio	0 = off, 1:1 … 20:1
atk	Attack	0.1 … 100 ms
dec	Release	**aut** = a 1/16 note of the tempo, else 20 … 1000 ms

FX3, PAGE 2: EQ

Displayed name	Name	Description
lo	Low shelf	100 Hz, –15 … +15 dB (64 = 0 dB; a band at 0 dB is off)
f1 g1 q1	Mid 1	frequency 20 Hz … 16 kHz, gain –15 … +15 dB, Q 0.3 … 8 (default 300 Hz, Q 1)
f2 g2 q2	Mid 2	as mid 1 (default 3 kHz)
hi	High shelf	8 kHz, –15 … +15 dB

FX3, PAGE 3

Displayed name	Name	Description
ord	Order	SCE (default), CSE, ESC, ECS, SEC, CES
amu	Auto gain	**off** · **fix**: half the reduction of a 0 dBFS peak · **ada**: the average reduction (≈ 1.5 s)
gai	Gain	make-up gain 0 … +24 dB, on top of **amu**
d/w	Dry/Wet	compressor mix (parallel compression)
bia	Sat bias	0 = symmetric; higher: even harmonics on loud signals

NOTES

- With **sat**, **rat** and **gai** at 0 and all EQ gains at 0 dB, FX3 is off and the sum stays unchanged.
- With the band split, **amu**, **gai** and **d/w** act on the compressed band only.
- **amu** = **ada** keeps compressed and dry signal about as loud: **d/w** then only changes the character.
- **bia** only acts with **sat** above 0. Use it moderately (20 … 50) and with **frq** at 150 … 300 Hz for a full mix: even harmonics of the bass make it muddy.
- **sat** and **frq** are also on the performance page.

EDIT VIEW

Top left: the peak levels of St1 / St2 before FX3 in dBFS (**-99** = silent).
Bottom left: **byp** = FX3 off, **on-4A** = on, highest gain reduction 4 dB, on St1 (**B** = following St1 to the St2 jacks).

Saturation and compressor together use about 5 % of the audio processing time.

7 > GLOBAL SETTINGS

SHIFT + CONFIG, second page (turn the encoder on). Like all global settings they are kept after a restart only when saved: SAVE → Global (or with a project save).

Displayed name	Name	Description
lim	Limiter OE	a limiter on each output after a 32 bit sum
ips	Instant pattern	chapter 4.4
mrv	Morph voices	kit morph on/off for the active voice (select it with its voice button); the edit view shows the morphed voices, e.g. **––56**
pre	Pre-fader sends	chapter 6.3

8 > MIDI

Numbers computed from the parameter list (NRPN n = parameter 128 + n), not measured on the device.

Message	Parameter
CC 1	kit morph **mrp** (global channel)
CC 2–5, 7, 8	OSC **wav**, voice 1–6
CC 122–127	LFO **mod**, voice 1–6
NRPN 27–32	LFO **wav**, voice 1–6
NRPN 57–62	LFO **ofs**, voice 1–6
NRPN 87–92	MIX **out**, voice 1–6
NRPN 93, 94, 97	OSC **len**, voice 1–3
NRPN 98	engines OE
NRPN 99	choke group **chk**
NRPN 100–105	OSC **sta**, voice 1–6
NRPN 106–117	FX1 OE

A > WHERE SETTINGS ARE STORED

Stored in	Settings
Kit	engine OE, **wav** (sample / wave number), **sta**, **len**, **chk**, **sna**, **snb**, **out**, LFO waveforms and **ofs**; FX1, **ord** of the FX slots; all of FX3
Pattern	step grid, flam values, trigger tracks, LFO start, velocity 0 steps
Global	FX2, **sya**, **syb**, the input gains (SHIFT + knob 1 / 2), **pre**, **ips**, **mrv**, **vsw**, **lim** OE
Project	the voice INIT template, the morph kit's values
Flash	samples and waves (imported from the card)
Not saved	**vmv**, **vmo**, **dj**, **res** OE

MORE INFORMATION

The complete description, with more examples and the history of each version, is in the repository:

- **README**: overview of all features and the installation.
- **docs/SAMPLES.md**: user guide of all additions.
- **docs/OVERVIEW.md** / **OVERVIEW.de.md**: one-page overview in English and German.
- **docs/ENGINES.md**: the voice engines OE.

github.com/shellbeat-jpg/lxr-desktop-sample

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